

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0821

Acec Grant 138 kV
August 22, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0821. The Interconnection Customer proposes a 188.6 MW natural gas combined cycle facility interconnecting at the Acec Grant 138 kV substation in the Maryland area. The request was received on July 04, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 353 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,885,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Brandon Shores to Broadland 138 kV
Contingency: Warren to Dubois 138 kV
Loading: 177.0% (Rating: 227 MVA)
DFAX: 17.5%
Pre-project loading: 162.4%
2. Facility: Brandon Shores to Orchard St 138 kV
Contingency: Dubois to Warren 138 kV
Loading: 173.3% (Rating: 221 MVA)
DFAX: 17.5%
Pre-project loading: 158.3%
3. Facility: Westport to Burlington Ks 138 kV L
Contingency: Dubois Substation to Warren 138 kV
Loading: 172.5% (Rating: 255 MVA)
DFAX: 17.4%
Pre-project loading: 159.7%
4. Facility: Warren to Dubois 138 kV
Contingency: Westport to Burlington Ks Substation 138 kV line
Loading: 169.8% (Rating: 206 MVA)
DFAX: 16.9%
Pre-project loading: 154.4%
5. Facility: Burlington Ks to Greene Street Station 138 kV
Contingency: Dubois to Warren 138 kV L
Loading: 154.1% (Rating: 232 MVA)
DFAX: 17.5%
Pre-project loading: 139.9%

6. Facility: Herbert A Wagner Substation to West Side 138 kV
Contingency: West Side to Wagner Substation 138 kV Line
Loading: 147.7% (Rating: 207 MVA)
DFAX: 11.7%
Pre-project loading: 137.1%
7. Facility: Orchard St Substation to Greene St Station 138 kV
Contingency: Warren to Dubois 138 kV
Loading: 145.3% (Rating: 240 MVA)
DFAX: 17.5%
Pre-project loading: 131.5%
8. Facility: W Side to Wgner 138 kV
Contingency: Herbert A Wagner to W Side 138 kV
Loading: 136.2% (Rating: 221 MVA)
DFAX: 11.7%
Pre-project loading: 126.2%
9. Facility: Dubois to Harlee Branch 138 kV
Contingency: Anatolia to Washington County 138 kV L
Loading: 130.9% (Rating: 170 MVA)
DFAX: 10.3%
Pre-project loading: 119.5%
10. Facility: Anatolia to Pumphrey 138 kV
Contingency: Anatolia to Washington County 138 kV line
Loading: 127.1% (Rating: 213 MVA)
DFAX: 10.3%
Pre-project loading: 118.0%
11. Facility: Pumphrey to Harlee Branch 138 kV
Contingency: Anatolia to Washington County 138 kV
Loading: 126.8% (Rating: 194 MVA)
DFAX: 10.3%
Pre-project loading: 116.8%
12. Facility: Washington County to Anatolia 138 kV
Contingency: Pumphrey to Anatolia 138 kV
Loading: 124.7% (Rating: 218 MVA)
DFAX: 10.3%
Pre-project loading: 115.8%
13. Facility: Dubois to Washington County 138 kV
Contingency: Pumphrey to Anatolia 138 kV
Loading: 121.9% (Rating: 185 MVA)
DFAX: 10.3%
Pre-project loading: 111.3%
14. Facility: Coldspring to Center 138 kV
Contingency: Hidalgo Smelter 345/1kV Transformer Bank
Loading: 109.5% (Rating: 275 MVA)
DFAX: 6.4%
Pre-project loading: 105.1%
15. Facility: Acec Grant to Acadia 138 kV L
Contingency: Sierra Pacific Lumber 138/1kV Transformer Bank
Loading: 101.0% (Rating: 193 MVA)
DFAX: 26.2%
Pre-project loading: 75.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brandon Shores to Broadland 138 kV	Rebuild line with higher-capacity conductor	\$564,000	100% (\$564,000)
Orchard St to Brandon Shores 138 kV L	Rebuild line with higher-capacity conductor	\$5,337,000	100% (\$5,337,000)
Westport to Burlington Ks 138 kV	Rebuild line with higher-capacity conductor	\$1,457,000	100% (\$1,457,000)
Warren to Dubois 138 kV	Rebuild line with higher-capacity conductor	\$1,372,000	100% (\$1,372,000)
Burlington Ks to Greene Street Station 138 kV	Rebuild line with higher-capacity conductor	\$359,000	100% (\$359,000)
Herbert A Wagner Substation to West Side 138 kV	Rebuild line with higher-capacity conductor	\$1,714,000	100% (\$1,714,000)
Orchard St to Greene Street Station Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,120,000	100% (\$1,120,000)
W Side to Wgner 138 kV	Rebuild line with higher-capacity conductor	\$2,489,000	100% (\$2,489,000)
Dubois to Harlee Branch 138 kV	Rebuild line with higher-capacity conductor	\$3,752,000	100% (\$3,752,000)
Anatolia to Pumphrey 138 kV	Rebuild line with higher-capacity conductor	\$767,000	100% (\$767,000)
Pumphrey to Harlee Branch 138 kV	Rebuild line with higher-capacity conductor	\$3,489,000	100% (\$3,489,000)
Washington County to Anatolia 138 kV	Rebuild line with higher-capacity conductor	\$1,788,000	100% (\$1,788,000)
Dubois to Washington County 138 kV line	Rebuild line with higher-capacity conductor	\$2,513,000	100% (\$2,513,000)
Coldspring to Center 138 kV	Reconductor line	\$620,000	100% (\$620,000)
Acec Grant to Acadia 138 kV L	Reconductor line	\$544,000	100% (\$544,000)
	Total allocated to project		\$27,885,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.